CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 18

16 BIT ADDITION USING 8085

AIM: To write and execute an 8085 assembly language program to find 16 bit addition and store result in memory.

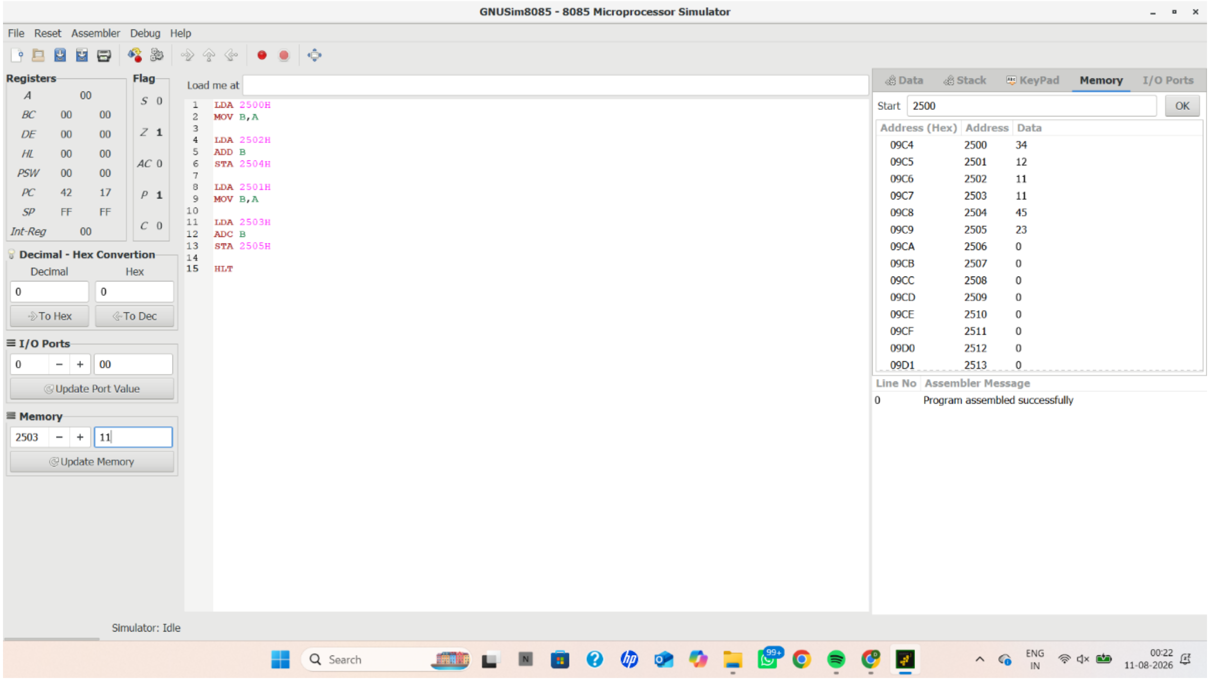
ALGORITHM

1. Load the lower byte of the first number into the accumulator.
2. Add the lower byte of the second number.
3. Store the result.
4. Load the higher bytes.
5. Add them along with the carry using ADC.
6. Store the higher-byte result.
7. Stop the program.

CODE:

```
LDA 2050H
MOV B,A
LDA 2052H
ADD B
STA 2054H
LDA 2051H
MOV B,A
LDA 2053H
ADC B
STA 2055H
MVI A,00H
ADC A
STA 2056H
HLT
```

OUTPUT:


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
Flag
Load me at
LDA 2500H
MOV B,A
LDA 2502H
ADD B
STA 2504H
LDA 2501H
MOV B,A
LDA 2503H
ADC B
STA 2505H
HLT
Data Stack KeyPad Memory I/O Ports
Start 2500 OK
Address (Hex) Address Data
09C4 2500 34
09C5 2501 12
09C6 2502 11
09C7 2503 11
09C8 2504 45
09C9 2505 23
Decimal - Hex Convertion
To Hex
To Dec
I/O Ports
Update Port Value
Memory
Update Memory
Line No Assembler Message
0 Program assembled successfully
Simulator: Idle